Evaluate Machine Translation using Automatic Metrics

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Introduction

- Machine Translation (MT) automates text conversion across languages while preserving meaning.
- India's linguistic diversity requires robust MT for digital accessibility, education, and governance.
- Challenges: grammatical differences, complex morphology, flexible word order.
- **Goal:** Evaluate pretrained MT models for Indian languages using automatic metrics and embedding similarity.

Problem Statement & Objectives

Problem:

- MT systems produce varying accuracy; evaluation is difficult.
- Indian languages pose extra challenges (morphology, word order, script variations).
- Manual evaluation is expensive; automatic metrics must capture semantic quality.

Objectives:

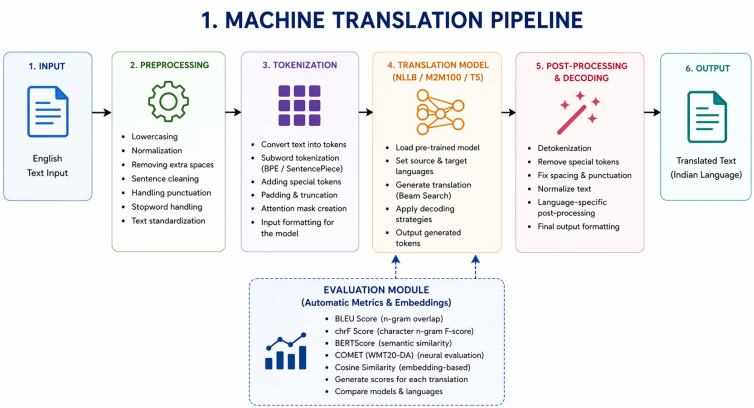
1. Analyze multilingual MT dataset.
2. Preprocess text for consistent evaluation.
3. Evaluate multiple pretrained models.
4. Compute BLEU, chrF, BERTScore, COMET, and cosine similarity.
5. Compare performance across languages.
6. Deploy a user-friendly web interface.

Dataset: IndicMTEval (AI4Bharat)

Feature	Description
Source	English sentences
Reference	Human translations
Prediction	MT-generated translations
Human Scores	MQM, Direct Assessment (DA)
Error Spans	Annotated translation errors

Languages evaluated: Hindi, Tamil, Malayalam, Marathi, Gujarati

Methodology: MT Pipeline



System Architecture

Models Used

Model	Description
NLLB-200 (600M distilled)	Meta AI, 200+ languages, encoder-decoder
M2M100	Direct translation without English pivot
T5-Base	Text-to-text transformer (prompt-based translation)
WMT20 COMET-DA	Neural evaluation metric

Automatic Evaluation Metrics

Metric	Description
BLEU	n-gram overlap with reference
chrF	Character n-gram F-score
BERTScore	Semantic similarity via BERT embeddings
COMET	Neural metric using source, reference, hypothesis
Cosine Similarity	Embedding similarity score

Results: Model Comparison

Model	BLEU	chrF	BERTScore	COMET	Cosine Sim
NLLB-200	0.142	41.30	0.618	0.596	0.731
M2M100	0.098	34.70	0.581	0.440	0.694
T5-Base	0.011	12.40	0.401	0.212	0.512

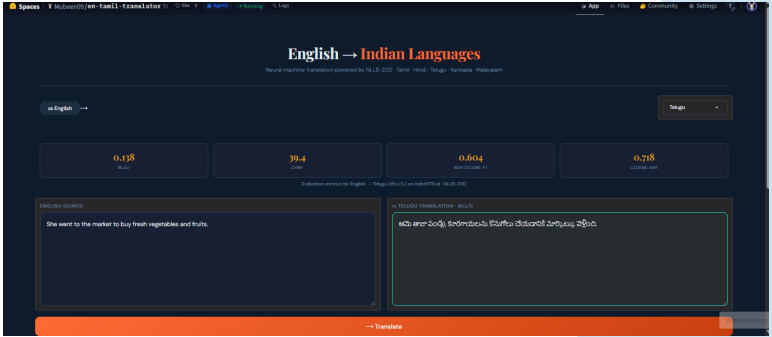
Observation: NLLB-200 outperforms others across all metrics.

Cross-Language Evaluation (NLLB-200)

Language	BLEU	chrF	COMET	Ensemble Score
Hindi	31.77	37.05	0.440	47.20
Tamil	8.455	58.68	0.596	49.29
Malayalam	11.95	60.29	0.561	50.38
Marathi	37.04	78.40	0.440	62.57
Gujarati	21.03	56.23	0.583	52.41

Insight: Marathi achieves highest lexical overlap; Tamil/Malayalam have better semantic scores.

Deployment: Web Interface



Features:

- Built with Gradio
- Hosted on Hugging Face Spaces
- Supports English → Tamil, Hindi, Telugu, Kannada, Malayalam
- Displays BLEU, chrF, BERTScore, cosine similarity in real time

Deployment Pipeline

1. Load pretrained `facebook/nllb-200-distilled-600M`
2. Local testing with Gradio: `python app.py`
3. Push code to GitHub repository
4. Connect GitHub to Hugging Face Spaces
5. Automatic deployment $\rightarrow$ public URL

Advantages:

- No local installation required
- Real-time translation
- Scalable cloud hosting

Conclusion

- Transformer-based multilingual models (especially NLLB-200) produce meaningful translations for Indian languages.
- Automatic metrics (BLEU, chrF, BERTScore, COMET) provide complementary insights into translation quality.
- Deployed system demonstrates practical usability for English $\rightarrow$ Indian languages translation.
- Semantic metrics correlate better with human judgment than lexical-only metrics.

Future Work

- **Expand language coverage:** Beyond Indian languages to global pairs (English-French, English-Arabic, etc.)
- **Fine-tuning:** Domain-specific adaptation (healthcare, legal, technical)
- **Advanced evaluation:** Incorporate BLEURT, human Direct Assessment (DA)
- **Scalable deployment:** Cloud-based inference with high request volume
- **New architectures:** Mixture-of-Experts (MoE), larger multilingual models

References

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Thank You

Questions?